NTNU

Physics-Based Modeling

From first principles to ODEs and PDEs — and how to solve them

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Outline

Physics-based modeling: the recipe

PDE benchmark: heat diffusion

PDE benchmark: Navier–Stokes

ODE benchmark: surface vessel

ODE benchmark: wind turbine

Synthesis: strengths, limits, and what comes next

01

Physics-based modeling: the recipe

What is physics-based modeling?

A model built *top-down* from conservation laws and constitutive relations — not fitted from data.

We trust

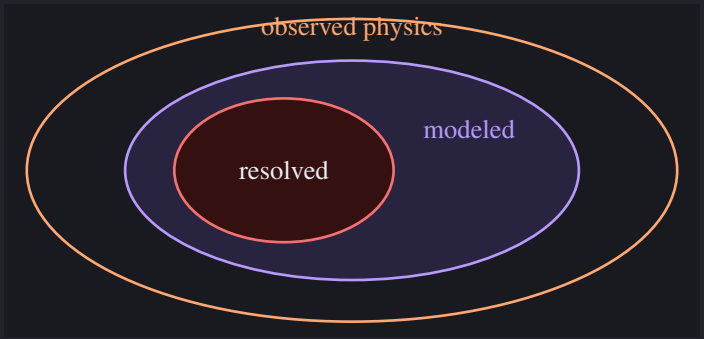
- Conservation of mass, momentum, energy
- Newton's laws, thermodynamics
- Well-tested constitutive laws (Fourier, Newtonian stress)

We must supply

- Material/hydrodynamic coefficients
- Initial & boundary conditions
- Closures for unresolved physics

Output: an ODE or PDE you can analyze and solve.

Resolved, modeled, observed physics



We can only *resolve* a fraction of reality; the rest is *modeled*, and a still larger part is merely *observed*. PBM lives in the inner two.

How we study each system: derive, discretize, results

Every benchmark in this lecture follows the **same three-part study** — and each part carries one lesson to keep in mind:

1. **Derive** the equation from first principles — a balance over a control volume or body.
   Lesson: exact conservation, but closed with *assumptions and modeled* terms.
2. **Discretize** it into a scheme that a computer can march.
   Lesson: further *simplifications* to make it solvable in reasonable time.
3. **Results** — run it, read the fields, find the weaknesses.
   Lesson: demanding but interpretable; you can see where it breaks.

Why repeat it four times

The structure is method-agnostic. *Derive* returns unchanged in every later module; *discretize* and the cost it exposes are exactly what data-driven and hybrid methods replace.

Four running benchmarks — reused all course long

Benchmark	Type	Governing physics
Heat diffusion	PDE	energy balance + Fourier's law
Navier–Stokes	PDE	mass + momentum conservation
Surface vessel	ODE	Newton–Euler rigid-body dynamics
Wind turbine	ODE	rotor angular-momentum balance

The course thesis

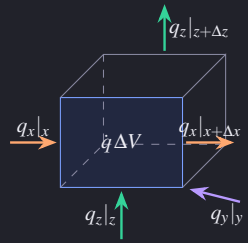
These same four problems are re-solved later with regression, neural networks, PINNs, operator learning, CoSTA and reduced-order models. Physics today is the *ground truth* everything else is judged against.

02

PDE benchmark: heat diffusion

Step 1a — Derive: energy balance on a control volume

A fixed box $\Delta V = \Delta x \Delta y \Delta z$. First law: the rate of internal-energy change equals net heat conducted in + generation.



First law on the box

$$\frac{\partial}{\partial t}(\rho c_p T)\,\Delta V = (\text{cond. in} - \text{out}) + \dot{q}\Delta V$$

Net outflow through the x-faces is $(q_x|_{x+\Delta x} - q_x|_x)\Delta y \Delta z$, and by Taylor $q_x|_{x+\Delta x} = q_x|_x + \dfrac{\partial q_x}{\partial x}\Delta x + \ldots$, so it is $\dfrac{\partial q_x}{\partial x}\Delta V$. The y- and z-faces give the same with q_y, q_z.

Divide by ΔV, let $\Delta V \to 0$ (truncation terms vanish); the balance is *exact*, written out term by term:

$$\rho c_p \frac{\partial T}{\partial t} = -\left(\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z}\right) + \dot{q}.$$

Step 1b — Derive: Fourier's law → the heat equation

Two unknowns remain — the temperature T and the three flux components. Close with **Fourier's law** (heat flows down the temperature gradient), component by component — the one *modeled* step:

$$q_x = -k\frac{\partial T}{\partial x}, \qquad q_y = -k\frac{\partial T}{\partial y}, \qquad q_z = -k\frac{\partial T}{\partial z}.$$

Substitute into the balance; with constant k each flux derivative becomes a second derivative of T:

Heat (diffusion) equation $(\alpha = k/\rho c_p)$

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) + \frac{\dot{q}}{\rho c_p}.$$

The balance is exact; Fourier's law is the empirical closure — the place data could enter later.

Step 2 — Write & analyze: IC, BC, scales

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) + \frac{\dot{q}}{\rho c_p}, \qquad T(\boldsymbol{x}, 0) = T_0(\boldsymbol{x}).$$

Boundary conditions

- Dirichlet: $T = T_b$ (fixed temperature)
- Neumann: $-k\,\partial_n T = q_n$ (fixed flux)
- Robin: $-k\,\partial_n T = h(T - T_\infty)$ (convection)

Analysis

- Linear, parabolic, smoothing
- Diffusion timescale $\tau \sim L^2/\alpha$
- Steady state: $\partial_x^2 T + \partial_y^2 T = -\dot{q}/k$

The diffusion time L^2/α tells you *how long* to simulate before you even discretize.

Step 3a — Discretize: Taylor in time and space

Grid $T_{i,j}^n \approx T(x_i, y_j, t^n)$. **Forward** difference in time (from $T(t+\Delta t) = T + \Delta t\, \partial_t T + \dots$):

$$\left.\frac{\partial T}{\partial t}\right|_{i,j}^{n} = \frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t} + \mathcal{O}(\Delta t).$$

Central 2nd difference in space (add the $\pm\Delta x$ expansions — odd terms cancel):

$$\left.\frac{\partial^2 T}{\partial x^2}\right|_{i,j}^{n} = \frac{T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n}}{\Delta x^2} + \mathcal{O}(\Delta x^2),$$

and the same in y. *First order* in time, *second order* in space.

Step 3b — Solve: the explicit FTCS update

Substitute both differences into $\partial_t T = \alpha(\partial_x^2 T + \partial_y^2 T) + \dot{q}/\rho c_p$ and solve for the only unknown, $T_{i,j}^{n+1}$:

FTCS update (forward-time, centred-space)

$$T_{i,j}^{n+1} = T_{i,j}^{n} + \alpha\Delta t\left[\frac{T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n}}{\Delta x^2} + \frac{T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n}}{\Delta y^2}\right] + \Delta t\,\frac{\dot{q}}{\rho c_p}.$$

Explicit: the right-hand side uses only known level-n values — every node updates independently, no linear system to solve.

But "no solve" is not free: the step Δt is capped, as the next slide shows. Notebook: `heat_diffusion.ipynb`.

Step 3c — Stability: von Neumann analysis

Why is Δt capped? Track one Fourier error mode $T_i^n = G^n e^{\mathrm{i}kx_i}$ through the 1-D update ($r = \alpha\Delta t/\Delta x^2$). Substituting and dividing out $G^n e^{\mathrm{i}kx_i}$:

$$G = 1 + r\left(e^{\mathrm{i}k\Delta x} - 2 + e^{-\mathrm{i}k\Delta x}\right) = 1 - 4r\sin^2\left(\frac{k\Delta x}{2}\right).$$

The mode stays bounded iff $|G| \le 1$ for *every* k, i.e. $r \le \frac{1}{2}$.

von Neumann stability limit (FTCS)

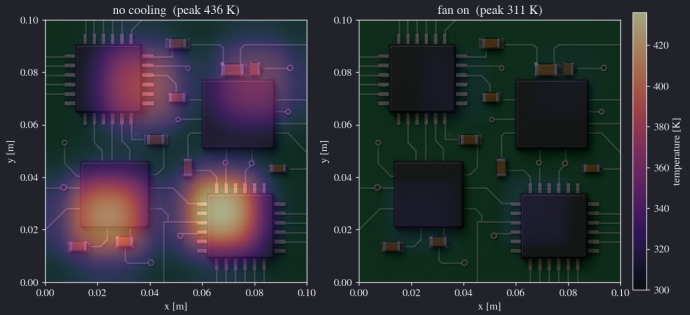
$$\Delta t \;\le\; \frac{1}{2\alpha}\cdot\frac{\Delta x^2\,\Delta y^2}{\Delta x^2 + \Delta y^2} \qquad \left(\text{1-D: } \Delta t \le \tfrac{\Delta x^2}{2\alpha}\right).$$

Halving Δx *quarters* Δt: the quadratic cost that motivates implicit schemes — and bites hardest on the stiff cable next.

Results 1 — a real circuit board, with and without a fan

Four chips heat a 10 cm board (higher conductivity under the chips); the source $\dot{q}$ lives only where a chip sits. Run the 2-D FTCS scheme to a quasi-steady state, then switch on a **fan** — a convective sink $-h(T - T_\infty)$ over the whole board.

Setup: 120×120 grid, 10 cm board, insulated edges; $\alpha_{\text{board}} = 10^{-5}$, $\alpha_{\text{chip}} = 5 \times 10^{-5}$ m^2/s, chip sources $\dot{q} \in \{55, 40, 70, 30\}$; $T_\infty = 300$ K, fan $h : 0 \to 6$ W/m^2/K.



Without cooling the hottest chip reaches ≈ 436 K and its heat bleeds toward its neighbours; the fan pulls the peak down to ≈ 311 K — an interpretable map from layout and cooling to hot-spot that updates in seconds.

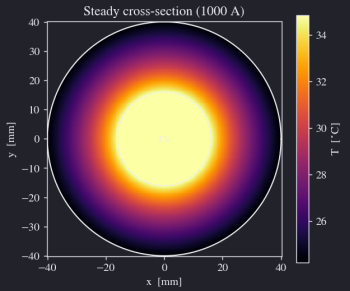
Worked example: the wind-farm export cable

A copper conductor carries current I ashore; ohmic loss I^2R' heats it, the XLPE insulation resists the flow, seawater carries it away. *How hot does the core run — and how much current is safe?*

Radial heat equation (per unit length)

$$\rho c_p \frac{\partial T}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(k r \frac{\partial T}{\partial r}\right) + \dot{q}(r)$$

Core $r < a$: $\dot{q} = I^2R'/(\pi a^2)$; insulation $a < r < b$: $\dot{q} = 0$.
Symmetry at $r = 0$; convection $-k\,\partial_n T = h(T - T_\infty)$ at $r = b$.
Setup: $a = 16$, $b = 40$ mm, $I = 1000$ A, $h = 20$ W/m^2/K, $T_\infty = 20°$C; backward Euler $\Delta t = 5$ s, $N = 160$ cells.

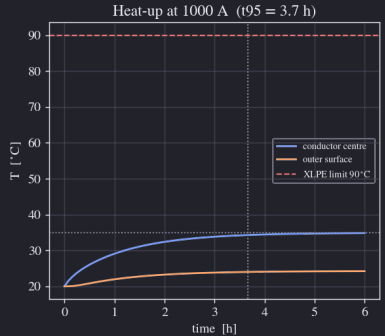


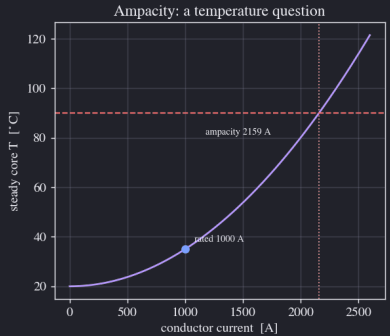
Our first *stiff* problem

Copper's α forces the explicit limit to $\Delta t \lesssim 0.3$ ms, yet the cable responds over *hours* — so we step with **backward Euler**. A first hint of why we later want reduced models.

Cable results: heating, ampacity, dynamic rating

Setup: default cable, $I = 1000$ A held to steady; backward Euler $\Delta t = 5$ s, $N = 160$ cells, integrated to 6 h.





Rated 1000 A $\Rightarrow$ core ≈ 35°C (55°C below the 90°C XLPE limit), reached only after $t_{95} \approx 3.7$ h; the limit is hit at an **ampacity of** ≈ 2160 **A**. The hours-long time constant is *headroom* — short overloads stay safe, the basis of **dynamic rating**.

Hand-off: the cooling h and temperature-dependent $R'(T)$ are uncertain — the source/BC terms a learned correction repairs (CoSTA, module 20), turning a static rating into a live twin.

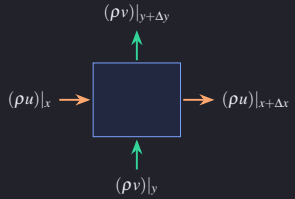
03

PDE benchmark: Navier–Stokes

Step 1a — Derive: mass conservation on a 2-D cell

Problem: find the velocity $\boldsymbol{u} = (u, v)$ and pressure p of an incompressible flow fixed by its walls and forcing.

A fixed cell $\Delta x \Delta y$ (unit depth). Rate of mass change = mass in − mass out; mass is carried across the x-faces by u, the y-faces by v.



$$\frac{\partial \rho}{\partial t}\Delta x \Delta y = \left[(\rho u)|_x - (\rho u)|_{x+\Delta x}\right]\Delta y + \left[(\rho v)|_y - (\rho v)|_{y+\Delta y}\right]\Delta x$$

Taylor each pair, divide by $\Delta x \Delta y$:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} = 0.$$

Incompressible limit (ρ constant)

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Not an evolution equation but a *constraint* on (u, v) at every instant — the structural fact that shapes the whole numerical method.

Step 1b — Derive: x- and y-momentum on the cell

Newton on the same cell for the x-momentum ρu: rate of change = net x-momentum carried in through the faces + pressure and viscous forces. x-momentum rides across the x-faces on u and the y-faces on v:

$$\frac{\partial(\rho u)}{\partial t}+\frac{\partial(\rho\, u\, u)}{\partial x}+\frac{\partial(\rho\, v\, u)}{\partial y}=-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right).$$

Expand the flux derivatives by the product rule and cancel using continuity $\partial_x u+\partial_y v=0$:

x- and y-momentum (component Navier–Stokes)

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\right)=-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right),$$

$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}\right)=-\frac{\partial p}{\partial y}+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right).$$

Nonlinear convection $u\,\partial_x u+v\,\partial_y u$ — the hard part; pressure carries no ∂_t: it is fixed by the constraint, not marched.

Step 2 — Write & analyze: pressure, Reynolds, vorticity

Pressure Poisson equation

Differentiate the two momentum equations and add, using $\partial_x u + \partial_y v = 0$:

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = -\rho \left[(\partial_x u)^2 + 2\,\partial_y u\,\partial_x v + (\partial_y v)^2 \right].$$

Pressure is fixed *instantly* by the velocity.

Vorticity ($\omega = \partial_x v - \partial_y u$)

$$\frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + v\frac{\partial \omega}{\partial y} = \nu\left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2}\right).$$

Pressure eliminated entirely.

The one dimensionless number

Reynolds number $\mathrm{Re} = UL/\nu$ = inertia / viscosity. Large Re $\Rightarrow$ thin layers, fine grids, possible turbulence.

Step 3a — Discretize: the projection (fractional-step) method

Central differences for every derivative on the grid $u^n_{i,j}, v^n_{i,j}, p_{i,j}$; split each step so velocity and the constraint decouple:

1. **Predict** (momentum without pressure):

$$u^* = u^n + \Delta t\big[-(u^n\,\partial_x u^n + v^n\,\partial_y u^n) + \nu(\partial_x^2 u^n + \partial_y^2 u^n)\big]$$

   and the same for v^*; in general $\partial_x u^* + \partial_y v^* \neq 0$.

2. **Pressure solve** — the 5-point Poisson stencil for p:

$$\frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta x^2} + \frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta y^2} = \frac{\rho}{\Delta t}\left(\partial_x u^* + \partial_y v^*\right),$$

   solved by iteration — it *couples every node*.

3. **Correct**: $u^{n+1} = u^* - \frac{\Delta t}{\rho}\partial_x p$, $v^{n+1} = v^* - \frac{\Delta t}{\rho}\partial_y p$, now with $\partial_x u^{n+1} + \partial_y v^{n+1} = 0$.

Step 3b — Stability and cost: CFL, and the Poisson bottleneck

The same Fourier-mode argument as heat, now on advection $\partial_t u + a\,\partial_x u = 0$ (Courant number $C = a\Delta t/\Delta x$): $|G| \le 1$ for all k requires $C \le 1$.

CFL condition

$$\Delta t \;\le\; \frac{\Delta x}{|\boldsymbol{u}|} \qquad (\text{Courant } C \le 1).$$

The bottleneck

A pressure Poisson solve *every* step — the dominant cost of incompressible CFD.

Notebooks

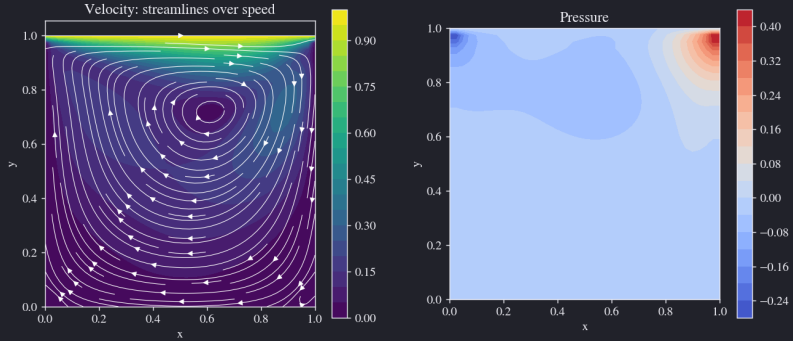
`navier_stokes_cavity.ipynb`
`navier_stokes_vortex.ipynb`

The convective cousin of the diffusive limit — both cap the explicit step by the same argument.

Results 1 — the lid-driven cavity

A moving top lid drags the fluid; the projection solver runs to steady state at $\mathrm{Re} = 100$.

Setup: 41×41 grid, unit square, lid $U = 1$, $\mathrm{Re} = 100$; projection with 50 Jacobi pressure iterations/step, $\Delta t = 0.0125$, 400 steps.

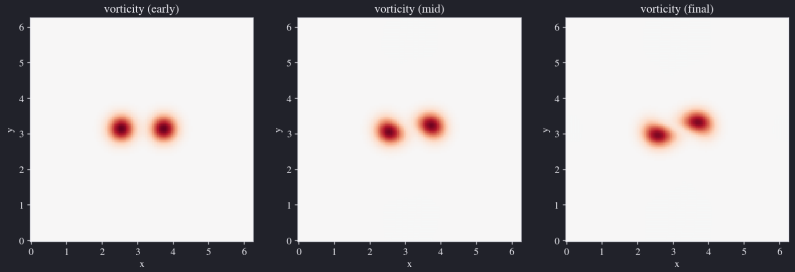


One primary vortex fills the cavity, its centre pushed toward the moving lid; the pressure (right) spikes at the downstream top corner. This flow has published reference profiles — the standard way to *validate* an incompressible solver.

Results 2 — vortex merger, and where PBM struggles

Two like-signed vortices in free space (vorticity form, $\mathrm{Re} = 2000$):

Setup: 128×128 periodic box $[0, 2\pi]^2$, $\mathrm{Re} = 2000$; spectral (FFT) Poisson solve, RK4 with $\Delta t = 10^{-3}$.



They rotate about each other and *merge* into a single vortex — a clean test of whether the solver conserves circulation as it should.

Where PBM struggles

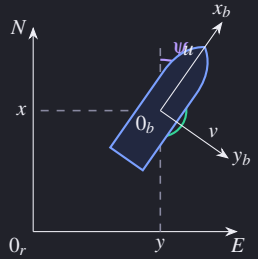
High Re and 3-D ⇒ huge grids, a Poisson solve every step, tiny time steps. Resolving real turbulence directly is unaffordable — the economic case for the reduced and neural-operator surrogates later.

04

ODE benchmark: surface vessel

Step 1a — Derive: Newton–Euler in scalar components

Problem: predict how a ship moves — its path, turning and overshoot — under thruster commands. A rigid hull in the horizontal plane, 3 DOF, in the body frame: surge u (forward), sway v (sideways), yaw rate r. Because the frame rotates with the ship, Newton's laws pick up the Coriolis cross-terms $-vr, +ur$:



Momentum & angular-momentum balance

$$m(\dot{u} - vr) = X, \quad m(\dot{v} + ur) = Y, \quad I_z \dot{r} = N$$

X, Y: total surge/sway force; N: yaw moment; m: mass; I_z: yaw inertia. The body origin 0_b sits at North–East position (x, y); the **heading** ψ is the angle from North to the bow x_b. The pose in the fixed frame then follows from

$$\dot{x} = u\cos\psi - v\sin\psi, \quad \dot{y} = u\sin\psi + v\cos\psi, \quad \dot{\psi} = r.$$

The forces X, Y, N collect the thrust $\boldsymbol{\tau}$ and the water's reaction — which we model next.

Step 1b — Hydrodynamic forces → the equations of motion

The water's reaction is three modeled effects (amber = not known exactly, from towing-tank tests):

- **Added mass** — the hull drags water as it accelerates, so the effective inertia grows: $m_{11} = m - X_{\dot{u}}$, $m_{22} = m - Y_{\dot{v}}$, $m_{33} = I_z - N_{\dot{r}}$.
- **Damping** — linear + quadratic drag opposing motion; **Coriolis** — the vr, ur cross-terms.

Component equations of motion (what the solver integrates)

$$\dot{u} = \left[\, \tau_1 + m_{22}\, vr - (d_u + X_{u|u|}|u|)\, u \,\right] / m_{11}$$
$$\dot{v} = \left[\, \tau_2 - m_{11}\, ur - (d_v + Y_{v|v|}|v|)\, v \,\right] / m_{22}$$
$$\dot{r} = \left[\, \tau_3 - (m_{22} - m_{11})\, uv - (d_r + N_{r|r|}|r|)\, r \,\right] / m_{33}$$

Compact shorthand for the same three equations: $\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}$, with $\boldsymbol{\nu} = [u, v, r]^\top$ (Fossen form).

Step 2 — Write & analyze: state space and stability

Stack into first-order state form $\dot{z} = f(z, \boldsymbol{\tau})$ with $z = [\boldsymbol{\eta}^\top, \boldsymbol{\nu}^\top]^\top \in \mathbb{R}^6$:

$$\dot{z} = \begin{bmatrix} \mathbf{R}(\psi)\,\boldsymbol{\nu} \\ \mathbf{M}^{-1}\big(\boldsymbol{\tau} - \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} - \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu}\big) \end{bmatrix}.$$

Equilibrium & linearization

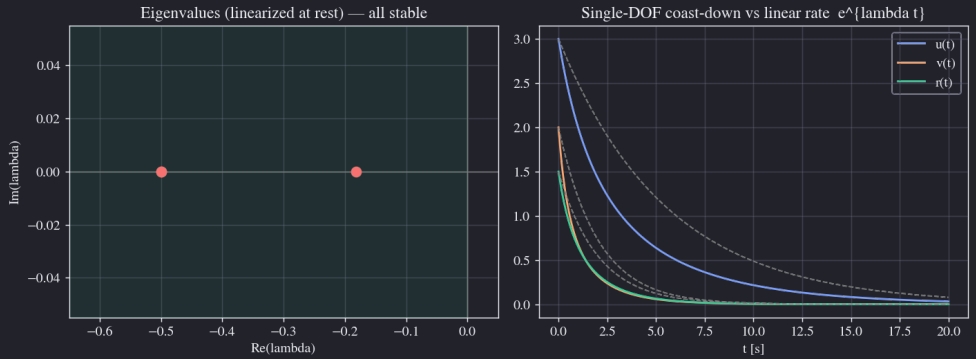
Steady motion $\dot{\boldsymbol{\nu}} = \mathbf{0}$; linearize the velocity dynamics: $\mathbf{M}\,\dot{\boldsymbol{\nu}} = -(\mathbf{C}_0 + \mathbf{D}_0)\boldsymbol{\nu}$.

Stability test

Eigenvalues of $-\mathbf{M}^{-1}(\mathbf{C}_0 + \mathbf{D}_0)$ must have negative real part. (The ODE analogue of von Neumann stability later.)

Nondimensionalize with length L, speed U (Froude scaling) to expose the few governing ratios.

Step 2 — Stability, demonstrated



Setup: thrust off; single-DOF coast-downs from $u_0 = 3$, $v_0 = 2$ m/s, $r_0 = 1.5$ rad/s over 20 s (RK4). *Left:* the velocity-dynamics eigenvalues all sit in the left half-plane — asymptotically stable. *Right:* with the thrust off, each velocity coasts to rest; the quadratic drag makes the early decay faster than the linear rate $e^{\lambda t}$ (dashed), relaxing onto it as the speed drops (`vessel_ode.ipynb`).

Step 3 — Discretize: time integration of the IVP

Integrate $\dot{z} = f(z, \tau(t))$ from $z(0) = z_0$. The base step is Taylor/Euler, $z_{n+1} = z_n + \Delta t f(z_n, t_n)$; **RK4** improves it by sampling the slope four times per step:

Explicit Runge–Kutta (RK4): four sampled slopes

$$k_1 = f(z_n,\ t_n), \qquad k_2 = f\left(z_n + \tfrac{\Delta t}{2} k_1,\ t_n + \tfrac{\Delta t}{2}\right),$$
$$k_3 = f\left(z_n + \tfrac{\Delta t}{2} k_2,\ t_n + \tfrac{\Delta t}{2}\right), \qquad k_4 = f\left(z_n + \Delta t\, k_3,\ t_n + \Delta t\right),$$

$$z_{n+1} = z_n + \tfrac{\Delta t}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right), \qquad \text{local error } \mathcal{O}(\Delta t^5).$$

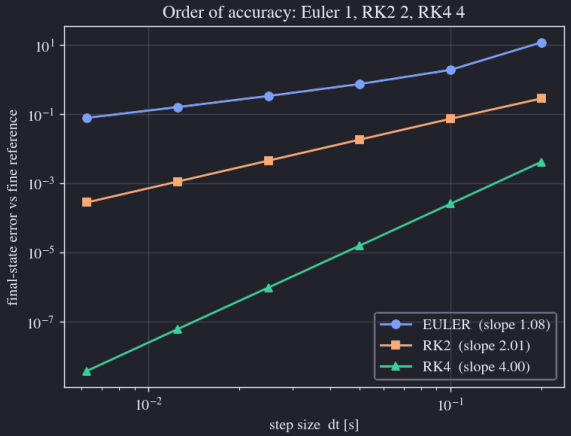
Step Δt chosen from *accuracy*, not stability (this ODE is non-stiff) — unlike the diffusion limit.

Caution

Stiff thruster steps $\rightarrow$ smaller Δt or an implicit integrator.

Notebook: `vessel_ode.ipynb`.

Step 3 — Order of accuracy, demonstrated



- Error vs step Δt against a fine reference.
- Fitted slopes: Euler ≈ 1, RK2 ≈ 2, **RK4** ≈ 4.
- At a fixed Δt, RK4 is orders of magnitude more accurate.

Why it matters

A single RK4 run is an *honest* ground truth: its discretization error is negligible next to the surrogate/PINN/operator errors measured against it later.

Setup: turning-circle input from rest, $t \in [0, 10]$ s; steps $\Delta t \in \{0.2, \ldots, 0.00625\}$ s vs an RK4 $\Delta t = 10^{-4}$ reference.

Step 4 — Analyze: trajectories and sensitivity

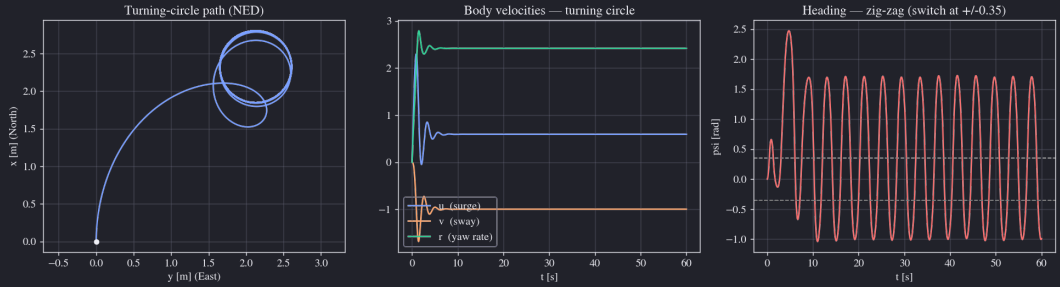
- Plot the path (x, y) and body velocities u, v, r (next slide).
- **Sensitivity**: perturb added mass / damping; watch the turning radius and overshoot change.

Where PBM struggles

The trajectory is only as good as the hydrodynamic coefficients, which drift with loading and fouling.

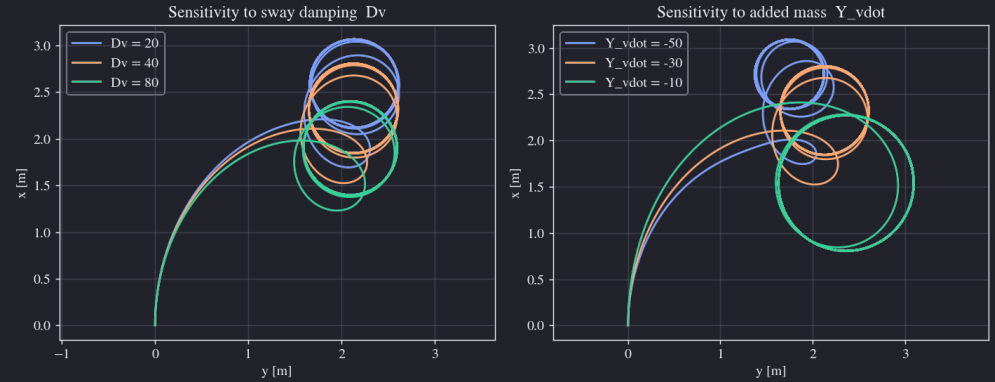
Hand-off: those uncertain coefficients are what the data-driven and hybrid-model modules identify or correct from data.

Step 4 — Trajectories from the RK4 solver



Setup: RK4 $\Delta t = 0.02$ s to 60 s; turning-circle thrust $\boldsymbol{\tau} = [200, 0, 80]$, zig-zag $\tau_3 = \pm 120$ switching at $\psi = \pm 0.35$ rad. *Left:* the turning circle spirals out of rest onto a constant-radius arc. *Middle:* the body velocities u, v, r settle to steady values. *Right:* the zig-zag heading overshoots each ± 0.35 switch line — yaw inertia carries it past before the reversed moment bites (`vessel_ode.ipynb`).

Step 4 — Sensitivity to the amber coefficients



Setup: same turning circle; $D_v \in \{20, 40, 80\}$ (def. 40), $Y_{\dot{v}} \in \{-50, -30, -10\}$ (def. -30), one at a time. Sweeping the uncertain sway damping D_v (left) and sway added mass $Y_{\dot{v}}$ (right) visibly shifts the turning circle. These are the "amber" coefficients — known only approximately — and precisely what later modules identify, correct, or rediscover (`vessel_ode.ipynb`).

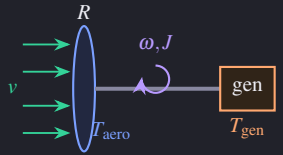
05

ODE benchmark: wind turbine

Step 1 — Derive: rotor angular-momentum balance

Problem: how fast should the rotor spin to capture the most power, and how do we cap it in strong wind?

A utility-scale turbine (NREL 5-MW). The drivetrain is one spinning shaft; Newton for rotation, $J\dot{\omega} = \sum$ torques, balances the driving aerodynamic torque against the braking generator torque:



Rotor dynamics (one state, ω)

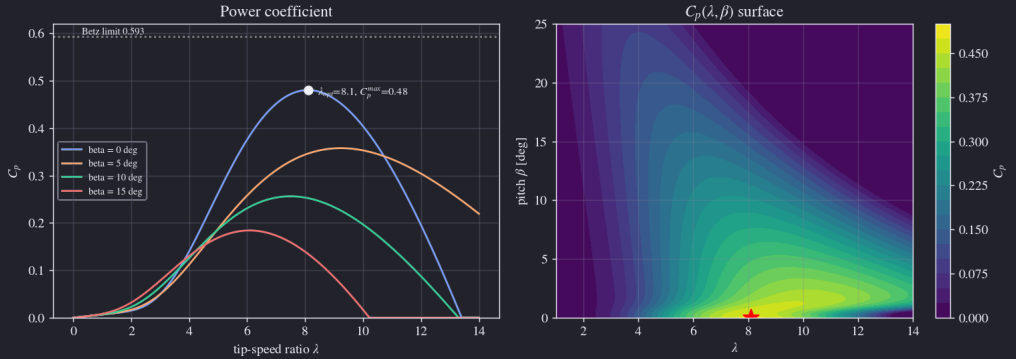
$$J\,\dot{\omega} = T_{\text{aero}} - T_{\text{gen}}$$

The aerodynamic torque is the captured power over the rotor speed:

$$T_{\text{aero}} = \frac{\frac{1}{2}\rho A\, C_p(\lambda,\beta)\, v^3}{\omega}, \qquad \lambda = \frac{\omega R}{v}.$$

λ: tip-speed ratio; β: pitch; $C_p(\lambda,\beta) \le$ Betz 0.593 is the amber aerodynamics — from blade-element theory / wind-tunnel data, never exact.

Step 1 — The power coefficient $C_p(\lambda, \beta)$



Setup: Heier C_p; left curves at pitch $\beta \in \{0, 5, 10, 15\}^\circ$, right surface over $\lambda \in [1, 14]$, $\beta \in [0, 25]^\circ$. *Left:* C_p versus tip-speed ratio for several pitch angles — a single optimum $\lambda_{\text{opt}} \approx 8.1$, $C_p^{\max} \approx 0.48$ at zero pitch; pitching lowers it. *Right:* the full $C_p(\lambda, \beta)$ surface, the operating optimum starred (`wind_turbine.ipynb`).

Step 2 — Discretize: integrating the rotor ODE

One scalar state ω — but each step hides an *algebraic loop*, because the torque depends on ω through the aerodynamics. Base step (forward Euler):

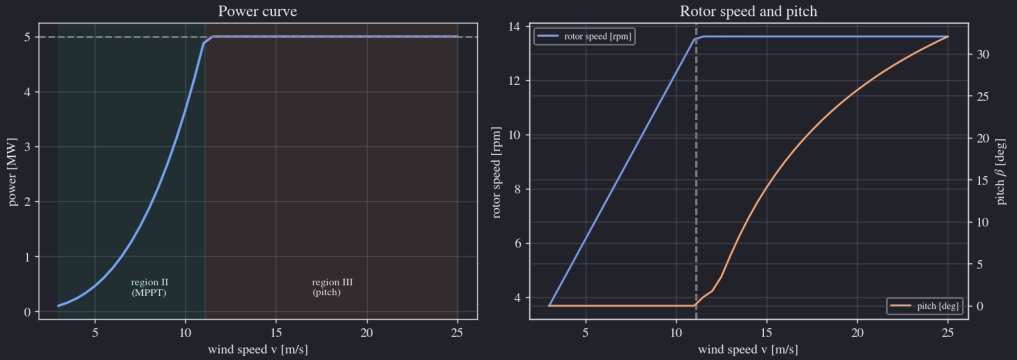
$$\omega_{n+1} = \omega_n + \Delta t \, \frac{T_{\text{aero}}(\omega_n, v_n) - T_{\text{gen}}(\omega_n)}{J}.$$

The per-step loop: evaluating the right-hand side

$$\omega \xrightarrow{\lambda = \omega R / v} \lambda \xrightarrow{\text{table}} C_p(\lambda, \beta) \longrightarrow T_{\text{aero}} = \frac{\frac{1}{2}\rho A C_p v^3}{\omega}, \qquad T_{\text{gen}} = K\omega^2.$$

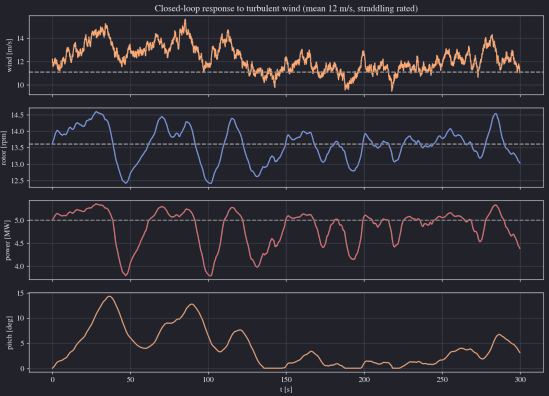
The same **RK4** four-stage step is used for accuracy, evaluating this loop at every stage. Step chosen from accuracy (non-stiff) — except the region-III pitch switch, which can stiffen the loop and force a smaller Δt.

Steps 3 — Control and the power curve



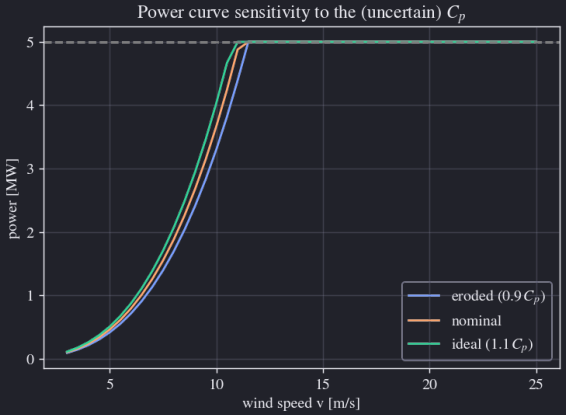
Setup: steady sweep $v \in [3, 25]$ m/s (cut-in to cut-out); $v_{\text{rated}} = 11.1$ m/s, rated 13.6 rpm, pitch PI $K_p = 18$, $K_i = 6$. Baseline strategy: **region II** torque control $T_{\text{gen}} = K\omega^2$ tracks λ_{opt} for maximum power (MPPT, C_p held at its peak); **region III** saturates the torque and uses *pitch* to hold rated speed and cap power at 5 MW (`wind_turbine.ipynb`).

Step 4 — Response to turbulent wind



Setup: Ornstein–Uhlenbeck wind, mean 12 m/s, TI = 12%, correlation time $\tau = 20$ s, 300 s at $\Delta t = 0.05$ s. Closed loop under realistic turbulence (mean 12 m/s, straddling rated): MPPT in lulls (pitch idle), pitch feathering in gusts to cap power — a seamless switch between regimes (`wind_turbine.ipynb`).

Sensitivity to the uncertain aerodynamics



Setup: $C_p \to sC_p$, $s \in \{0.90, 1.0, 1.10\}$; mean power $\{2.68, 2.86, 3.02\}$ MW. Blade erosion, fouling or an imperfect aerodynamic model scale C_p; below rated, captured power scales with it — the wind analogue of the vessel's amber coefficients (`wind_turbine.ipynb`).

06

Synthesis: strengths, limits, and what comes next

What physics-based modeling gives us — and doesn't

Strengths

- Interpretable, every term has meaning
- Extrapolates on physical grounds
- Needs no training data
- Respects conservation by construction

Limits

- Expensive (Poisson solves, tiny steps)
- Unknown coefficients / closures
- Unmodeled physics, wrong source terms
- Calibration data is scarce

The gap

Every limit on the right is a place where *data* can help — without throwing away the physics on the left.

The three lessons of this lecture

We derived, discretized and ran four models. Three things should stick:

1. First principles — but built on assumptions

Conservation of mass, momentum and energy is *exact*. Everything that *closed* the equations is a model: Fourier's law, the Newtonian stress, lumped added-mass and damping, the C_p surface — the amber terms we circled all lecture.

2. Simplified again to be solvable in reasonable time

Continuous equations became finite stencils; stability capped the step ($\Delta t \propto \Delta x^2$ for diffusion, $C \leq 1$ for advection); incompressibility forced a Poisson solve every step; 3-D was cut to 2-D.

3. Interpretable — but expensive, and weak where the amber lives

Every term has meaning and the RK4/FTCS references are honest ground truth — yet the cable needs sub-millisecond steps, CFD pays a Poisson solve per step, and outputs are visibly sensitive to coefficients we do not know precisely.

Making way for data: the same four benchmarks, many ways

Next: data-driven modelling — learn the model *from data* where deriving it falls short. The rest of the course patches the amber terms and the cost, without discarding the physics:

Method	What it changes
Data-driven modeling	fits the input–output map directly
Physics-informed NN	embeds the derived residual in the loss
Operator learning (DeepONet, FNO)	learns the solution operator cheaply
Corrective source term	adds a learned term to the physics
Equation discovery	recovers the governing law *from data*
Reduced-order modeling	compresses the expensive solver

Carry forward

The four solutions you build today — heat, Navier–Stokes, vessel, turbine — are the reference every one of these methods is judged against.

Summary

1. PBM is derived top-down from conservation laws and modeled closures.
2. One three-part study — *derive, discretize, results* — spans ODEs and PDEs alike.
3. Stability analysis appears twice: eigenvalues (ODE) and von Neumann/ CFL (PDE).
4. Four benchmarks — heat, Navier–Stokes, vessel, turbine — anchor the entire course.
5. PBM's limits (cost, unknown closures, scarce data) set up every data-driven and hybrid method to come.



NTNU

First principles first — then let data help

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